

0.1 Solution

The problem requires counting monic polynomials $P(x)$ of degree 7 with integer coefficients satisfying three conditions: all coefficients have odd absolute values strictly less than 2025^2 , the polynomial is monic, and the absolute value of every coefficient is a root of P .

0.1.1 Key Observations and Constraints

1. **Root Condition for Coefficients:** For each coefficient c_i , $|c_i|$ must be a root of P . Since P has degree 7, the set $S = \{|c_i| \mid i = 0, \dots, 7\}$ can have at most 7 distinct elements. However, there are 8 coefficients, so by the pigeonhole principle, at least two coefficients share the same absolute value.
2. **Leading Coefficient and Root 1:** The leading coefficient is 1, so $|1| = 1$ must be a root. Thus, $P(1) = 0$, implying the sum of all coefficients is zero. The leading term is 1, so the sum of the remaining 7 coefficients must be -1 .
3. **Odd Coefficients:** All coefficients must be odd integers, as their absolute values are odd.
4. **Bounding Coefficient Magnitudes:** All $|c_i| < 2025^2$, which is satisfied for small values like 1 and 3.

0.1.2 Case Analysis

Case 1: All Coefficients Are ± 1

- Here, $S = \{1\}$, so $P(1) = 0$ is the only root condition required.
- The sum of the 7 non-leading coefficients must be -1 . Let k be the number of $+1$ s among these coefficients. Then:

$$k - (7 - k) = -1 \implies 2k = 6 \implies k = 3.$$

- The number of ways to choose 3 positions for $+1$ s among 7 coefficients is $\binom{7}{3} = 35$.
- All such polynomials satisfy the conditions, as $|c_i| = 1$ is a root (via $P(1) = 0$) and coefficients are odd with magnitudes $< 2025^2$.

Case 2: Coefficients Include ± 3

- Let $S = \{1, 3\}$. Then $P(1) = 0$ and $P(3) = 0$ must hold.
- The leading coefficient is 1, so $P(3) = 0$ requires balancing the dominant 3^7 term. Analysis shows the coefficient of x^6 must be -3 (to cancel 3^7), and the remaining terms must satisfy:

$$\sum_{i=0}^5 a_i 3^i = 0 \quad \text{and} \quad \sum_{i=0}^6 a_i = -1.$$

where $a_i \in \{\pm 1, \pm 3\}$ and $a_6 = -3$.

- **Subcases Based on Constant Term:**

- **Subcase A:** Constant term $a_0 = 3$. Solving the system yields two valid polynomials:
 1. Coefficients: $[3, -1, 3, -1, -3, 1, -3]$ (for x^0 to x^6).
 2. Coefficients: $[3, -1, -3, 1, 3, -1, -3]$.
- **Subcase B:** Constant term $a_0 = -3$. Solving the system yields one valid polynomial:
 - * Coefficients: $[-3, 1, 3, -1, 3, -1, -3]$.

- These three polynomials satisfy all conditions: $P(1) = 0$, $P(3) = 0$, coefficients are odd with magnitudes $< 2025^2$, and all $|c_i| \in \{1, 3\}$ are roots.

0.1.3 Final Count

- **Case 1 contributes 35 polynomials.**
- **Case 2 contributes 3 polynomials.**
- No overlap exists between the cases, as Case 2 polynomials include ± 3 .

0.1.4 Conclusion

The total number of good polynomials is $35 + 3 = 38$.

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